

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

1. Evaluate each of the following integrals using IBP.

a) $\int (2x+1)\cosh(x)dx$

Let $u = 2x+1$ & $dv = \cosh(x)dx$ and $du = 2dx$ & $v = \sinh(x)$

$$\begin{aligned}\int (2x+1)\cosh(x)dx &= (2x+1)\sinh(x) - \int 2\sinh(x)dx \\ &= (2x+1)\sinh(x) - 2\cosh(x) + C\end{aligned}$$

b) $\int \frac{x}{\cos^2(x)}dx$

Let $u = x$ & $dv = \frac{1}{\cos^2(x)}dx$ and $du = dx$ & $v = \tan(x)$

$$\begin{aligned}\int \frac{x}{\cos^2(x)}dx &= x\tan(x) - \int \tan(x)dx \\ &= x\tan(x) - \int \frac{\sin(x)}{\cos(x)}dx\end{aligned}$$

Let $u = \cos(x)$, then $du = -\sin(x)dx \rightarrow -du = \sin(x)dx$. Therefore,

$$\begin{aligned}x\tan(x) - \int \frac{\sin(x)}{\cos(x)}dx &= x\tan(x) + \int \frac{1}{u}du \\ &= x\tan(x) + \ln|u| + C \\ &= x\tan(x) + \ln|\cos(x)| + C\end{aligned}$$

c) $\int_1^e \frac{\ln(x)}{\sqrt{x}} dx$

First note $\int_1^e \frac{\ln(x)}{\sqrt{x}} dx = \int_1^e x^{-1/2} \ln(x) dx$

Let $u = \ln(x)$ & $dv = x^{-1/2} dx$ and $du = \frac{1}{x} dx$ & $v = 2\sqrt{x}$

$$\int_1^e \frac{\ln(x)}{\sqrt{x}} dx = \int_1^e x^{-1/2} \ln(x) dx = 2 \ln(x) \sqrt{x} \Big|_1^e - \int_1^e 2\sqrt{x} \frac{1}{x} dx$$

$$= (2\sqrt{e} - 0) - 2 \int_1^e \frac{1}{\sqrt{x}} dx$$

$$= 2\sqrt{e} - 4\sqrt{x} \Big|_1^e$$

$$= 2\sqrt{e} - (4\sqrt{e} - 4)$$

$$= 4 - 2\sqrt{e}$$

2. On which of the following would be easy to use Tabular Integration by Parts?

- | | | |
|----------------------------------|-------------------------|--------------------------------------|
| (i) $\int x^5 \sin(\pi x) dx$ | (ii) $\int x \ln(x) dx$ | (iii) $\int \tan(x) \sin(x) dx$ |
| (iv) $\int (2^x + 1) \cos(x) dx$ | (v) $\int x^2 e^x dx$ | (vi) $\int (3x + 2) \tan^{-1}(x) dx$ |

Remember that in order to use Tabular Integration by Parts one of the factors in the integral needs to be a polynomial and the other factor needs to represent something that you could take several antiderivatives of.

3. Evaluate $\int x^5 \sin(\pi x) dx$.

We will proceed using tabular integration by parts.

$p(x)$	$F(x)$	Sign
x^5	$\sin(\pi x)$	
$5x^4$	$-\frac{1}{\pi} \cos(\pi x)$	+
$20x^3$	$-\frac{1}{\pi^2} \sin(\pi x)$	-
$60x^2$	$\frac{1}{\pi^3} \cos(\pi x)$	+
$120x$	$\frac{1}{\pi^4} \sin(\pi x)$	-
120	$-\frac{1}{\pi^5} \cos(\pi x)$	+
0	$-\frac{1}{\pi^6} \sin(\pi x)$	-

Therefore,

$$\int x^5 \sin(\pi x) dx = -\frac{1}{\pi} x^5 \cos(\pi x) + \frac{5}{\pi^2} x^4 \sin(\pi x) + \frac{20}{\pi^3} x^3 \cos(\pi x) - \frac{60}{\pi^4} x^2 \sin(\pi x) - \frac{120}{\pi^5} x \cos(\pi x) + \frac{120}{\pi^6} \sin(\pi x) = C$$

4. $\int x^3 e^{x^2} dx$ (Hint: Start with a substitution to make things easier)

$$\text{Let } t = x^2 \text{ \& } dt = 2x dx \rightarrow \frac{1}{2} dt = x dx.$$

$$\text{Therefore, we have } \int x^3 e^{x^2} dx = \int \frac{1}{2} t e^t dt$$

$$\text{Let } u = \frac{1}{2} t \text{ \& } dv = e^t dt \text{ and } du = \frac{1}{2} dt \text{ \& } v = e^t$$

$$\int x^3 e^{x^2} dx = \int \frac{1}{2} t e^t dt = \frac{1}{2} t e^t - \frac{1}{2} \int e^t dt = \frac{1}{2} t e^t - \frac{1}{2} e^t + C = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C$$

5. Evaluate the following integral $\int e^x [\cos(x) + \sin(x)] dx$.

Note: There are several ways to approach this question. The question actually is much easier if you reverse my selection for u and dv on the first integral and don't do anything with the 2nd integral. Can you see why? Another approach would be to note that if you multiply out the integrand it already resembles a derivative rule we know from calculus 1 and so you can immediately guess the correct antiderivative.

$$\int e^x [\cos(x) + \sin(x)] dx = \int e^x \cos(x) dx + \int e^x \sin(x) dx$$

We need to perform an integration by parts for each of these integrals.

For the first integral let $u = \cos(x)$ and $dv = e^x dx$ which implies that $du = -\sin(x) dx$ & $v = e^x$.

$$\text{So, } \int e^x \cos(x) dx = e^x \cos(x) + \int e^x \sin(x) dx$$

Then let $u = \sin(x)$ and $dv = e^x dx$ which implies that $du = \cos(x) dx$ & $v = e^x$.

Thus,

$$\begin{aligned} \int e^x \cos(x) dx &= e^x \cos(x) + \int e^x \sin(x) dx \\ &= e^x \cos(x) + \left[e^x \sin(x) - \int e^x \cos(x) dx \right] \\ \Rightarrow \int e^x \cos(x) dx &= e^x \cos(x) + e^x \sin(x) - \int e^x \cos(x) dx \\ 2 \int e^x \cos(x) dx &= e^x \cos(x) + e^x \sin(x) \\ \int e^x \cos(x) dx &= \frac{1}{2} [e^x \cos(x) + e^x \sin(x)] + C_1 \end{aligned}$$

For the second integral let $u = \sin(x)$ and $dv = e^x dx$ which implies that $du = \cos(x) dx$ & $v = e^x$.

$$\text{So, } \int e^x \sin(x) dx = e^x \sin(x) - \int e^x \cos(x) dx$$

Then let $u = \cos(x)$ and $dv = e^x dx$ which implies that $du = -\sin(x) dx$ & $v = e^x$.

Thus,

$$\begin{aligned} \int e^x \sin(x) dx &= e^x \sin(x) - \int e^x \cos(x) dx \\ &= e^x \sin(x) - \left[e^x \cos(x) + \int e^x \sin(x) dx \right] \\ \Rightarrow \int e^x \sin(x) dx &= e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x) dx \\ 2 \int e^x \sin(x) dx &= e^x \sin(x) - e^x \cos(x) \\ \int e^x \sin(x) dx &= \frac{1}{2} [e^x \sin(x) - e^x \cos(x)] + C_2 \end{aligned}$$

Therefore, we have

$$\begin{aligned} \int e^x [\cos(x) + \sin(x)] dx &= \int e^x \cos(x) dx + \int e^x \sin(x) dx \\ &= \frac{1}{2} [e^x \cos(x) + e^x \sin(x)] + C_1 + \frac{1}{2} [e^x \sin(x) - e^x \cos(x)] + C_2 \\ &= e^x \sin(x) + C \end{aligned}$$

6. When performing integration by parts we define $dv = g(x)dx$ and then compute $v = G(x)$. It is a frequent question to ask why we don't write $v = G(x) + C$. Show that in general $uv - \int v du = u(v + C_1) - \int (v + C_1) du$ thereby justifying the omission of the constant of integration when calculating v in integration by parts)

Note that

$$\begin{aligned} u(v + C_1) - \int (v + C_1) du &= uv + uC_1 - \int v du - \int C_1 du \\ &= uv + uC_1 - \int v du - C_1 u \\ &= uv - \int v du \end{aligned}$$

7. Use calculus I integration techniques (differentiation rules, algebra, & u-substitutions) to answer each of the following questions.

- a) Evaluate the integral $\int_0^1 (1-x^2)^9 x^9 dx$.

Let $u = 1 - x^2$, then $du = -2x dx \rightarrow -\frac{1}{2} du = x dx$ and $x^2 = 1 - u$.

The bounds of the integral become $u(0) = 1$ and $u(1) = 0$.

So we have

$$\begin{aligned} \int_0^1 (1-x^2)^9 x^9 dx &= \int_1^0 -\frac{1}{2} u^9 (1-u)^4 du = \frac{1}{2} \int_0^1 u^9 (1-4u+6u^2-4u^3+u^4) du \\ &= \frac{1}{2} \int_0^1 (u^9 - 4u^{10} + 6u^{11} - 4u^{12} + u^{13}) du \\ &= \frac{1}{2} \left(\frac{1}{10} u^{10} - \frac{4}{11} u^{11} + \frac{6}{12} u^{12} - \frac{4}{13} u^{13} + \frac{1}{14} u^{14} \right) \Big|_0^1 = \frac{1}{2} \left(\frac{1}{10} - \frac{4}{11} + \frac{1}{2} - \frac{4}{13} + \frac{1}{14} \right) \\ &= \frac{1}{20020} \end{aligned}$$

b) Suppose that $\int_{-\pi/2}^{\pi/2} [e^{\sin(x)} \cos(x)] dx = a - \frac{1}{a}$ where $a > 0$. Determine the value of a .

We calculate that $\int_{-\pi/2}^{\pi/2} [e^{\sin(x)} \cos(x)] dx = \int_{-1}^1 e^u du = e^1 - e^{-1} = e - \frac{1}{e}$. Therefore, $a = e$.

c) Evaluate the definite integral $\int \frac{dx}{1+e^{-x}}$.

$$\int \frac{dx}{1+e^{-x}} = \int \frac{dx}{1+\frac{1}{e^x}} = \int \frac{1}{\left(\frac{e^x+1}{e^x}\right)} dx = \int \frac{e^x}{1+e^x} dx$$

Let $u = e^x + 1$ & $du = e^x dx$. Then,

$$\int \frac{e^x}{1+e^x} dx = \int \frac{1}{u} du = \ln|u| + C = \ln(e^x + 1) + C$$

d) Let $a > 1$ be a constant satisfying the equation below. What is the value of $a^2 + a + 1$?

$$\int_1^a \frac{x-1}{x+\sqrt{x}} dx = 4$$

Note that

$$\begin{aligned} \int_1^a \frac{x-1}{x+\sqrt{x}} dx &= \int_1^a \frac{x + (\sqrt{x} - \sqrt{x}) - 1}{x + \sqrt{x}} dx = \int_1^a \frac{x + \sqrt{x}}{x + \sqrt{x}} dx - \int_1^a \frac{\sqrt{x} + 1}{x + \sqrt{x}} dx = \int_1^a 1 dx - \int_1^a \frac{\sqrt{x} + 1}{\sqrt{x}(\sqrt{x} + 1)} dx \\ &= \int_1^a 1 dx - \int_1^a \frac{1}{\sqrt{x}} dx = \int_1^a 1 dx - \int_1^a \frac{1}{\sqrt{x}} dx = (a-1) - 2\sqrt{x} \Big|_1^a = (a-1) - (2\sqrt{a} - 2) = a - 2\sqrt{a} + 1 \end{aligned}$$

Since $\int_1^a \frac{x-1}{x+\sqrt{x}} dx = 4$ it is the case that

$$a - 2\sqrt{a} + 1 = 4 \rightarrow a - 2\sqrt{a} - 3 = 0 \rightarrow (\sqrt{a} - 3)(\sqrt{a} + 1) = 0 \rightarrow \sqrt{a} = 3 \rightarrow a = 9$$

Therefore, $a^2 + a + 1 = 81 + 9 + 1 = 91$